Chapter test

16 Stimuli, receptors and responses

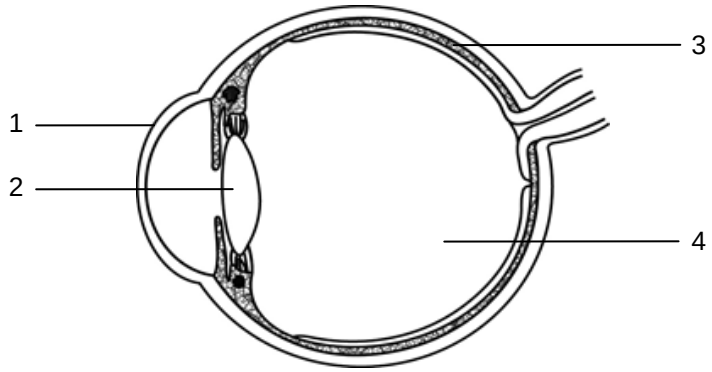
Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

Direction: Questions 1 and 2 refer to the diagram below, which shows a section of a human eye.



1. Which of the above labelled structures is / are involved in converging light rays onto the retina?

   A. 2 only
   B. 1 and 2 only
   C. 2 and 4 only
   D. 1, 2 and 4 only ☐

2. Which of the following statements about structure 3 is / are correct?

   (1) It supplies nutrients and oxygen to the cells of the eye.
   (2) It is shiny to reflect light.
   (3) It is tough in order to protect the eyeball.

   A. (1) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

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3. The curvature of the lens can be changed by the contraction or relaxation of the

   A. eye muscles.
   B. iris muscles.
   C. ciliary muscles.
   D. suspensory ligaments. ☐

4. What happens in the eye when a person walks from bright daylight into a dark room?

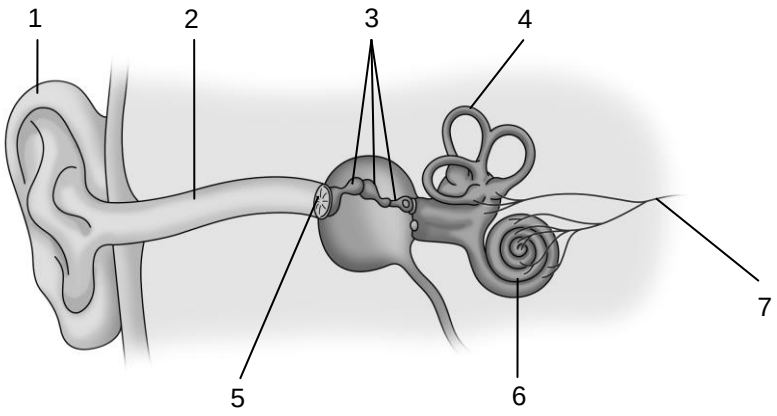
	Circular muscles of the iris	***Radial muscles of the iris***	***Pupil***
A.	contract	contract	constricts
B.	contract	relax	dilates
C.	relax	contract	dilates
D.	relax	relax	constricts

☐

★ 5. When people get old, they usually have difficulties in focusing their eyes when looking at nearby objects. This is because

   A. their cornea has become uneven.
   B. their lens has become inelastic.
   C. their vitreous humour has become cloudy.
   D. their eyeball has become smaller. ☐

Directions: Questions 6 to 8 refer to diagram below, which shows the structure of a human ear.



6. Which of the following structures is not involved in hearing?

   A. 4
   B. 5
   C. 6
   D. 7 ☐

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7. Structure 3 helps to

(1) amplify vibrations.
(2) transmit vibrations.
(3) equalize the pressure on both sides of structure 5.

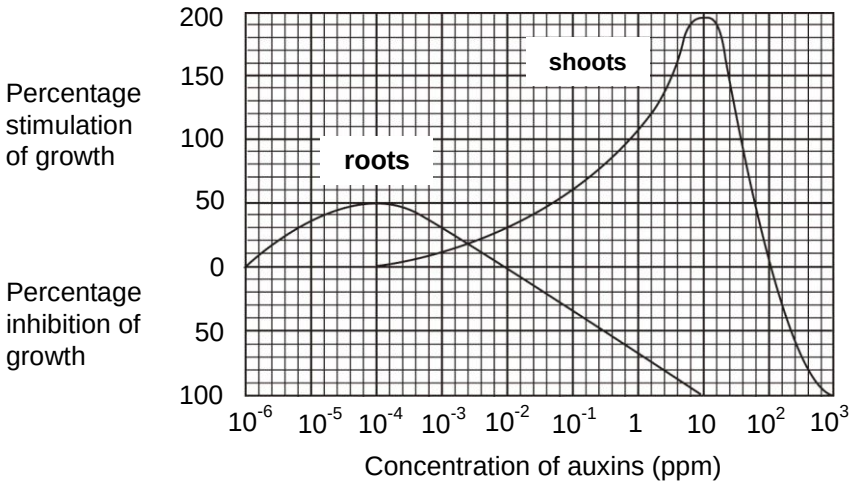
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★ 8. What would happen if structure 7 were cut?

(1) Hearing of that ear would be lost.
(2) Touch of structure 4 would not be felt.
(3) Maintenance of body balance would become difficult.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

Directions: Questions 9 to 10 refer to the graph below, which shows the effect of auxin concentration on the growth of roots and shoots.

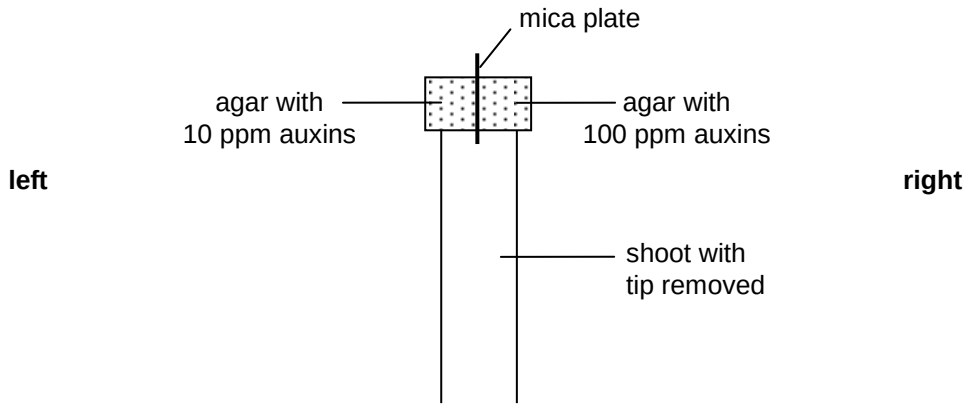


9. Which auxin concentrations will stimulate growth of both the roots and shoots?

A. 10^{-6} to 10^{-4} ppm
B. 10^{-4} to 10^{-2} ppm
C. 10^{-4} to 10^{2} ppm
D. 10^{-6} to 10^{2} ppm ☐

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★★ 10. Two pieces of agar with different concentrations of auxins are placed on top of a shoot with its tip removed as shown below.



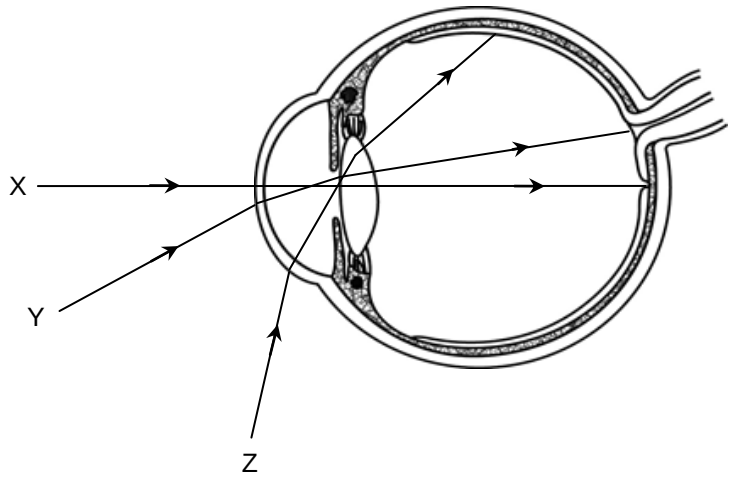
Predict the growth of the shoot.

A. It will grow towards the left.
B. It will grow towards the right.
C. It will grow straight upwards.
D. It will not grow at all. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	16.2	16.2	16.2	16.2	16.2	16.3	16.3	16.3	16.4	16.4

B. Problem solving (23 marks)

★ 11. The diagram below shows a section of a human eye and the light rays from a small object when it is placed at three different positions, X, Y, and Z. The eye keeps focusing in the direction of X.



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(a) Compare and explain the vision produced when the object is placed at positions X and Y. (4 marks)

(b) The object can be seen when it is placed at position Y, but its colour cannot be detected. Explain why. (3 marks)

(Total: 7 marks)

12. (a) A man was standing by the roadside and looking at a bus moving towards him. Describe and explain the changes that occurred in his eyes. (3 marks)

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(b) The man had an eye defect in which he was unable to see clearly the route number of bus when the bus was far away from him.

(i) State ***two*** features of the eyeball that might have caused this eye defect. (2 marks)

_____

_____

(ii) Draw a ray diagram to show how this eye defect can be corrected by wearing a suitable lens. (2 marks)

(c) The man had normal colour vision, but he found it difficult to distinguish colour in dim light. Explain why. (2 marks)

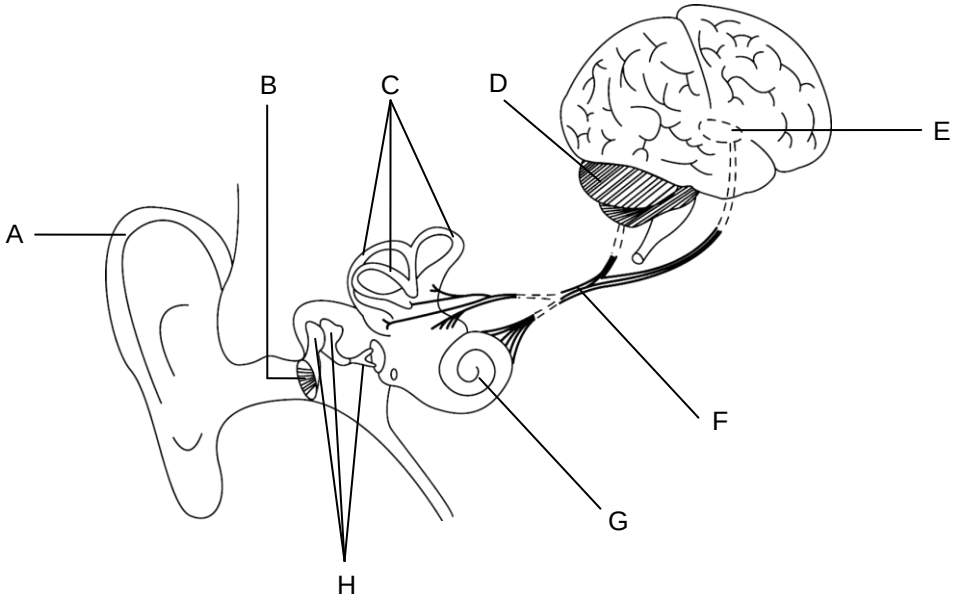
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_____

(Total: 9 marks)

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★ 13. The diagram below shows a human ear and its connection to the brain.



(a) Using the letters provided, state all the parts, in the correct sequence, involved in the process of hearing. (2 marks)

(b) Using the letters provided, identify the part(s) that

(i) convert(s) sound waves to mechanical vibrations. (1 mark)

(ii) contain(s) sensory hair cells. (1 mark)

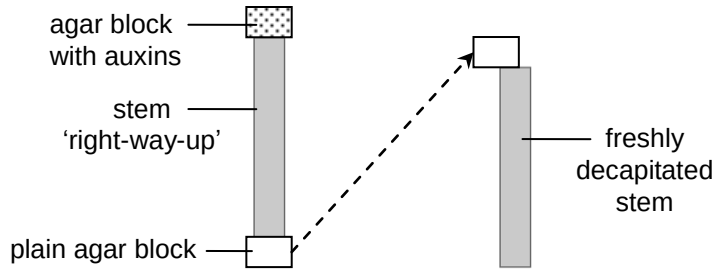
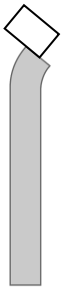
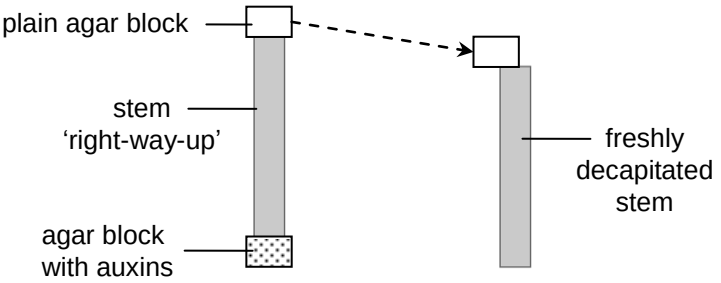
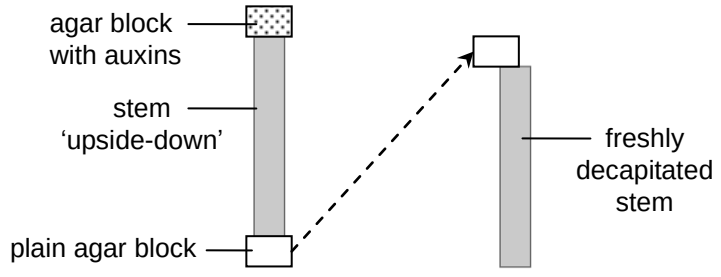
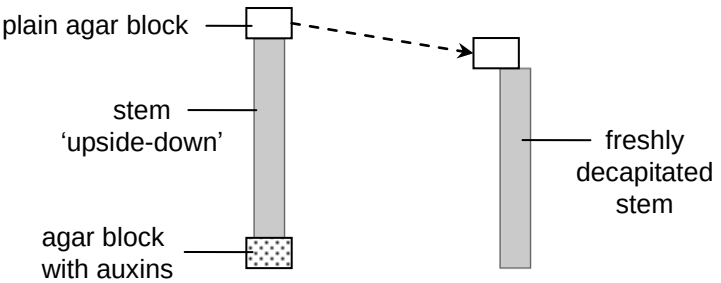
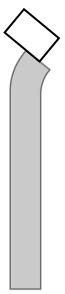
(c) Jenny took the lift from the ground floor to the observation deck on the 100th floor of a sightseeing tower. Explain why she could not hear clearly when she reached the observation deck. (3 marks)

(Total: 7 marks)

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C. Exam challenge (7 marks)

★★ 14. To investigate the movement of auxins in plants, a plain agar block and an agar block containing auxins were placed at different ends of a cut stem. The plain agar block acted as the 'receiving' block for any auxins that moved through the stem during a 24-hour period. The 'receiving' agar block was then placed on one side of a freshly decapitated stem. Any resulting bending of the stem was observed after two days. The diagram below shows the experimental set-ups and the results.

	Experimental set-up	Result
I		
II		
III		
IV		

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(a) Explain the result of set-up I. (3 marks)

(b) What conclusions about the movement of auxins can be drawn from the results of all four set-ups? Explain your answers. (4 marks)

(Total: 7 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	D	2.	A	3.	C	4.	C	5.	B
6.	A	7.	A	8.	B	9.	B	10.	B

(1 mark each)

B. Problem solving (23 marks)

11. (a) The object can be seen when it is at position X, but not at position Y. (1)
When the object is at position X, its image falls on the yellow spot. (1)
When it is at position Y, its image falls on the blind spot. (1)
The yellow spot has photoreceptors to detect the image but the blind spot has no photoreceptors. (1)

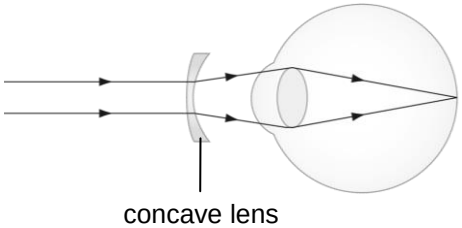
(b) When the object is at position Z, its image falls on the periphery of the retina (1)
where rod cells are present to detect light from the object. (1)
However, there are fewer cone cells, thus the colour of the object cannot be detected effectively. (1)

(Total: 7 marks)

12. (a) The ciliary muscles contracted. (1)
The tension in the suspensory ligament was reduced. (1)
Thus the lens became thicker / more convex due to its elasticity. (1)

(b) (i) The eyeball is too long. (1)
The lens is too thick. (1)

(ii) Correct type of lens (1)
Correct pathway of light rays (1)



(c) Cones, which are responsible for colour vision, (1)
cannot function properly in dim light. (1)

(Total: 9 marks)

13. (a) A → B → H → G → F → E (2)

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(b) (i) B (1)

(ii) C, G (1)

(c) On reaching the observation deck, the air pressure on the two sides of the eardrum became unequal. (1)

The eardrum was pressed outwards. (1)

Thus, it could not vibrate freely in response to sound vibrations. (1)

(Total: 7 marks)

C. Exam challenge (7 marks)

14. (a) Auxins moved down to the 'receiving' agar block. (1)

After the 'receiving' agar block was placed on the left side of the decapitated stem, auxins moved down the left side, (1)

causing the cells on the left side to elongate more /causing the left side to grow faster. (1)

Thus, the stem bent to the right.

(b) The movement of auxins is not simply by diffusion / Auxins do not always move from high to low concentration (1)

because when the stem was 'right-way-up' the auxins moved down into the 'receiving' block, but not up if auxins are provided from below. (1)

Auxins move through the stem only from the apical end (1)

because when the stem was 'upside-down' the auxins did not move down, but was able to move up if auxin was provided from below. (1)

(Total: 7 marks)

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